

Nonlinear Model Predictive Control of Flexible Ammonia Production

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Abstract

Flexible dynamic operation of ammonia synthesis is a key enabling factor of green ammonia production in the face of intermittency in renewable energy sources. A nonlinear model predictive control scheme is presented in this paper for the Haber-Bosch ammonia synthesis process with a varying feed flowrate. The considered process consists of three interstage-cooled reactor beds and a flash separator, with a flexible reactor load varying between 50% and 100% of its nominal capacity. The proposed control scheme aims to control the reactor temperatures, separation pressure, and the liquid volume in the flash tank during feed transitions. A simulation study is performed given an assumed 5-hour feed schedule. The results indicate that all controlled variables are maintained in a safe operating range and tracked with small offsets in the nominal case as well as under disturbances in the reaction rate. It takes approximately 10 minutes for the reactor temperatures and 40 minutes for the separation pressure to reach steady state after a large step change in the reactor load. In the comparison with a PID control scheme, the investigated NMPC scheme tracks feed temperatures and the separation pressure significantly faster with less oscillations. This study therefore demonstrates the feasibility and effectiveness of the proposed control scheme in enabling flexible operation of the process.

Keywords: process control, green ammonia, nonlinear model predictive control

1. Introduction

Ammonia, a crucial chemical substance, is the second most produced chemical globally with an annual production amount of 176 million tons (David et al., 2020). Approximately 88% of ammonia produced is used as a fertilizer in agriculture, underscoring its significant role in this sector (Pattabathula and Richardson, 2016). Virtually all ammonia is synthesized through the Haber-Bosch process, in which nitrogen is converted to ammonia through a reaction with hydrogen at a temperature of 300-500 °C and a pressure around 200 bar (Erisman et al., 2008).

Ammonia production currently relies heavily on fossil fuels. Specifically, over 90% of hydrogen used in ammonia production is sourced from natural gas reforming or coal gasification (Pattabathula and Richardson, 2016), contributing to around 2% of global CO₂ emissions (Smith and Torrente-Murciano, 2023). Replacing this hydrogen with one produced via water electrolysis powered by renewable energy sources (RES) will be a key step towards decarbonizing agriculture. Ammonia produced using such renewable hydrogen is referred to as green ammonia, and its production and utilization are attracting worldwide attention currently by industry, private investors, and academia (Wang et al., 2022). In addition to its role in decarbonizing agriculture, green ammonia has tremendous potential as a medium for storing renewable energy. Chemical-based energy storage (compared to batteries and mechanical energy storage systems) is a promising way to resolve large mismatches

between the supply and demand of renewable power and enable long duration storage and high storage capacity. Candidate chemicals for use as storage media include hydrogen, ammonia, methanol, and methane (Palys and Daoutidis, 2022). Ammonia has three times higher energy density than hydrogen and costs 10-100 times less to store than hydrogen (Salmon and Bañares-Alcántara, 2021). In addition, it is a carbon-free fuel. Due to these advantages, green ammonia has been proposed as an energy vector for many renewable energy systems (Du et al., 2022), such as sustainable agricultural systems (Palys et al., 2021), power distribution systems (Riley et al., 2023), microgrids (Kong et al., 2024), and energy hubs (Ghappani and Karimi, 2023).

Green ammonia production is likely to be powered by stand-alone renewable energy systems to ensure low carbon intensity in the electricity used. It will therefore be constrained by the availability of renewable power and consequently by the available inventory of hydrogen and nitrogen. Due to the intermittency of renewable power resources, ammonia production at a constant rate requires batteries and storage tanks to buffer fluctuations in the power supply and the feedstocks of hydrogen and nitrogen. Flexible ammonia production on the other hand, which largely follows the availability of RES, lends itself to significantly lower storage needs for hydrogen and/or power. An economic comparison between these two paradigms indicates that flexible operation leads to a significant reduction in the levelized cost of ammonia production (Palys and Daoutidis, 2023a) and is thus crucial for the economic viability of green ammonia production.

Flexible operation of air separation processes for nitrogen production has been well documented (Klein et al., 2021). Pro-

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adopted in this work, the nominal ammonia concentration is 97.65 mol%. The nominal liquid volume is set to 30% of the total volume of the flash tank, with an acceptable range between 25% and 40% of the total volume. The production rate can be manipulated by adjusting the valve at the bottom of the flash tank. Another valve is set at the outlet of the vapor phase to regulate the recycle flowrate. Also, a purge valve is deployed to regulate the tank pressure. According to the nominal flash pressure (196 bar) and the feed pressure of the reactor (200 bar), the operating pressure range is set to be between 190 and 200 bar.

In the modeling of the synthesis loop, some assumptions and simplifications are made. Compressors are not modeled; it is assumed that the streams fed to C1 and C2 can always be pressurized to the desired pressure. Also, the modeling of heat exchanger HE-2 is not considered. It is assumed that the cold utility used in HE-2 is sufficient to cool the hot stream to the nominal temperature (313 K).

In the scenario of our interest, the plant is connected to a fluctuating RES, and the feed flowrate schedule of hydrogen and nitrogen is specified at a supervisory level. It is assumed that the ratio of hydrogen and nitrogen in the feed stream is fixed at 3:1 on a molar basis. The feed flowrate is restricted to 50% - 100% of the nominal reactor capacity.

Table 1: Design parameters and operating conditions of the ammonia synthesis loop.

Parameter	Specification	Value
L_1	Length of the 1st reactor bed, m	1.24
L_2	Length of the 2nd reactor bed, m	1.79
L_3	Length of the 3rd reactor bed, m	2.83
A	Cross-section area of the reactor beds, m ²	1.13
ε	Bed void fraction	0.33
T_f	Temperature of the feed stream, K	523
P_{total}	Total pressure of the feed stream, bar	200
U_{EX}	Overall heat transfer coefficient of HE-1, W/m ² /K	536
A_{EX}	Heat transfer area of HE-1, W/m ² /K	180
T_{sep}	Temperature in the separator, K	313
V_{sep}	Volume of the separator, m ³	100

Table 2: Catalyst properties.

Parameter	Catalyst property	
	Type	Metallic iron on Fe ₃ O ₄
ρ_c	Density	2200 kg/m ³
Cp_c	Heat capacity	1100 J/kg/K
D_p	Particle hydraulic diameter	2.85 mm

2.1. Reactor Model

The reactor model is adopted from Morud and Skogestad (1998), in which the gas-phase holdup is neglected since the gas density is low. Each reactor bed is discretized in space to

n sections. Let $\mathcal{I} := \{1, 2, 3\}$ be the set of reactor beds and $\mathcal{J} := \{1, 2, \dots, n\}$ be the set of reactor sections. Then, i denotes the index of reactor beds and j denotes the index of reactor sections. The reaction rate (r_{ij}) is as follows:

$$r_{ij} = \frac{2\alpha}{\rho_c} \cdot \left(k_1 \frac{P_{N_2,ij} P_{H_2,ij}^{1.5}}{P_{NH_3,ij}} - k_{-1} \frac{P_{NH_3,ij}}{P_{H_2,ij}^{1.5}} \right), \forall i \in \mathcal{I}, j \in \mathcal{J} \quad (1)$$

$$k_1 = 1.79 \times 10^4 e^{-87090 \text{ J mol}^{-1} / RT_{c_{ij}}}, \forall i \in \mathcal{I}, j \in \mathcal{J} \quad (2)$$

$$k_{-1} = 2.75 \times 10^{16} e^{-198464 \text{ J mol}^{-1} / RT_{c_{ij}}}, \forall i \in \mathcal{I}, j \in \mathcal{J} \quad (3)$$

where ρ_c is the density of catalyst particles, R is the ideal gas constant, $T_{c_{ij}}$ is the catalyst temperature, and $P_{H_2,ij}$, $P_{N_2,ij}$, $P_{NH_3,ij}$ are the partial pressures of H₂, N₂, NH₃, respectively. The coefficient $\alpha = 4.75$ corrects the activity of the newer industrial catalyst (Morud and Skogestad, 1998). Let $\mathcal{S} := \{H_2, N_2, NH_3\}$ be the set of chemical species. The partial pressure of each component is given by Dalton's Law:

$$P_{sij} = P_{total} z_{sij}, \forall s \in \mathcal{S}, i \in \mathcal{I}, j \in \mathcal{J} \quad (4)$$

in which P_{total} is the total pressure and z_{sij} is the mole fraction of component s at each reactor section.

Let m_i denote the feed mass flowrate of reactor bed i , w_{sij} denote the mass fraction of species s at each reactor section, A be the cross-sectional area of the reactor bed, L_i be the total length of reactor bed i , and ε be the bed void fraction. The length of a single reactor section is given by $l_i = L_i/n$. The mass balance for ammonia is as follows:

$$m_i(w_{NH_3,ij} - w_{NH_3,ij-1}) = A l_i (1 - \varepsilon) \rho_c r_{ij}, \forall i \in \mathcal{I}, j \in \mathcal{J} \quad (5)$$

For hydrogen and nitrogen, mass balances are governed by the reaction stoichiometry:

$$\frac{m_i(w_{H_2,ij-1} - w_{H_2,ij})}{MW_{H_2}} = \frac{3}{2} \cdot \frac{m_i(w_{NH_3,ij} - w_{NH_3,ij-1})}{MW_{NH_3}}, \forall i \in \mathcal{I}, j \in \mathcal{J} \quad (6)$$

$$\frac{m_i(w_{N_2,ij-1} - w_{N_2,ij})}{MW_{N_2}} = \frac{1}{2} \cdot \frac{m_i(w_{NH_3,ij} - w_{NH_3,ij-1})}{MW_{NH_3}}, \forall i \in \mathcal{I}, j \in \mathcal{J} \quad (7)$$

where MW_{H_2} , MW_{N_2} , and MW_{NH_3} are the molecular weights of H₂, N₂, and NH₃, respectively.

Energy balance equations are derived from Wang et al. (2017, 2020), in which the heat transfer between the catalyst solid and the fluid is considered:

$$(1 - \varepsilon) \rho_c C p_c \frac{dT_{c_{ij}}}{dt} = -\Delta H_r r_{ij} (1 - \varepsilon) \rho_c - \dot{Q}_{ij}, \forall i \in \mathcal{I}, j \in \mathcal{J} \quad (8)$$

$$\varepsilon \rho_{ij} C p_{ij} \frac{dT_{g_{ij}}}{dt} = -\frac{m_i C p_{ij}}{A l_i} (T_{g_{ij}} - T_{g_{ij-1}}) + \dot{Q}_{ij}, \forall i \in \mathcal{I}, j \in \mathcal{J} \quad (9)$$

In the equations above, $\dot{Q}_{ij}$ is the heat transfer rate between the solid catalyst and the fluid, $T_{g_{ij}}$ is the fluid temperature, ΔH_r is the reaction enthalpy, ρ_{ij} is the gas density, and $C p_{ij}$

and C_{p_c} are specific heat capacities of the fluid and catalyst, respectively. $C_{p_{ij}}$ is given by:

$$C_{p_{ij}} = \sum_{s \in S} \bar{C}_{p_s} w_{sij}, \quad \forall i \in I, j \in J \quad (10)$$

where $\bar{C}_{p_s}$ is the specific heat capacity of species s . $\dot{Q}_{ij}$ is given as follows:

$$\dot{Q}_{ij} = U_{ij}(T_{c_{ij}} - T_{g_{ij}}), \quad \forall i \in I, j \in J \quad (11)$$

where U_{ij} is the heat transfer coefficient, given by the following equations:

$$Pr_{ij} = \frac{C_{p_{g,ij}} \mu}{\lambda}, \quad \forall i \in I, j \in J \quad (12)$$

$$Re_{ij} = \frac{D_p m_i}{A(1 - \varepsilon) \mu}, \quad \forall i \in I, j \in J \quad (13)$$

$$Nu_{ij} = (2 + 1.1 Re_{ij})^{0.6} (Pr_{ij})^{0.33}, \quad \forall i \in I, j \in J \quad (14)$$

$$U_{ij} = \frac{\lambda Nu_{ij}}{\pi D_p^2}, \quad \forall i \in I, j \in J \quad (15)$$

where Pr_{ij} is the Prandtl number, Re_{ij} is the Reynolds number, Nu_{ij} is the Nusselt number, μ is the dynamic viscosity, λ is the thermal conductivity, and D_p is the catalyst particle diameter.

2.2. Heat Exchanger Model

The heat exchanger HE-1 is modeled as a counter-current exchanger using the NTU method (Incropera et al., 1996). Let C_c denote the heat capacity rate (i.e., mass flow rate multiplied by the specific heat capacity) of the cold stream fed to HE-1, and C_h be the heat capacity rate of the hot stream (the output from the 3rd reactor bed). $T_{HEC,0}$ and $T_{HEC,1}$ represent the inlet and the outlet temperature of the cold stream, and $T_{HEH,0}$, $T_{HEH,1}$ represent the inlet and outlet temperature of the hot stream, respectively. The amount of heat transferred q is calculated by:

$$q = \epsilon C_{min}(T_{HEH,0} - T_{HEC,0}) \quad (16)$$

$$C_{min} = \min\{C_c, C_h\} \quad (17)$$

The effectiveness ϵ is a function of the number of transfer units (NTU) and heat capacity ratio C_r :

$$\epsilon = \frac{1 - e^{-NTU(1-C_r)}}{1 - C_r e^{-NTU(1-C_r)}} \quad (18)$$

$$NTU = \frac{U_{EX} A_{EX}}{C_{min}} \quad (19)$$

$$C_{max} = \max\{C_c, C_h\} \quad (20)$$

$$C_r = \frac{C_{min}}{C_{max}} \quad (21)$$

where U_{EX} is the overall heat transfer coefficient and A_{EX} is the heat transfer area. The outlet temperatures of the hot and cold streams are given as follows:

$$T_{HEH,1} = T_{HEH,0} - \frac{q}{C_h} \quad (22)$$

$$T_{HEC,1} = T_{HEC,0} + \frac{q}{C_c} \quad (23)$$

In addition, a minimum temperature difference for heat exchange ΔT is considered:

$$T_{HEH,0} - T_{HEC,0} \geq \Delta T \quad (24)$$

2.3. Separator Model

Vapor-liquid equilibrium (VLE) is assumed for the flash separation process: the distribution of ammonia in the vapor and liquid phase follows Raoult's Law, while Henry's Law is applied for the VLE calculations of hydrogen and nitrogen Gullberg (2018). Phase equilibrium equations are as follows:

$$P_{sep} y_{NH_3} = P_{NH_3}^{sat} x_{NH_3} \quad (25)$$

$$P_{sep} y_s = H_s x_s, \quad \forall s \in \{H_2, N_2\} \quad (26)$$

where P_{sep} is the pressure in the separator, y_s is the vapor mole fraction of species s , and x_s is the liquid mole fraction of species s . The Henry's constants H_s and the saturation pressure of ammonia $P_{NH_3}^{sat}$ are constant, as the flash separation is assumed to be isothermal.

The dynamic mass balances of the flash tank are as follows:

$$\frac{dN_s}{dt} = \frac{m_3 w_{s3n}}{MW_s} - F_v y_s - F_l x_s, \quad \forall s \in S \quad (27)$$

$$N_l x_s + N_v y_s = N_s, \quad \forall s \in S \quad (28)$$

where F_v and F_l are the outlet molar flowrates in the vapor and liquid phases of the flash tank, respectively; N_s is the holdup of species s in the flash tank, N_l and N_v are the liquid and gas holdups, respectively. In Eq. 27, the term $\frac{m_3 w_{s3n}}{MW_s}$ is the input molar flowrate of species s , and $F_v y_s$ and $F_l x_s$ are the output molar flowrates of species s in the vapor and liquid phase, respectively. The pressure is calculated by the ideal-gas law:

$$P_{sep} = \frac{N_v R T_{sep}}{V_{sep} - V_l} \quad (29)$$

where T_{sep} is the temperature in the separator, V_{sep} is the volume of the separator, and the liquid volume V_l is given by:

$$V_l = N_l \sum_{s \in S} \frac{x_s MW_s}{\rho_{l_s}} \quad (30)$$

where ρ_{l_s} is the density of species s in liquid.

2.4. Mixing and Splitting

The mass flowrate of the recycle stream m_{cyc} is as follows:

$$m_{cyc} = F_{cyc} \sum_{s \in S} y_s MW_s \quad (31)$$

Let F_p denote the purge flowrate; the recycle molar flowrate F_{cyc} is then given by:

$$F_{cyc} = F_v - F_p \quad (32)$$

The purge flowrate can not be greater than the vapor outlet flowrate of the flash tank:

$$F_p \leq F_v \quad (33)$$

The recycle stream is mixed with the feed stream and split into four streams. Let $\mathcal{V} := \{1, 2, 3, 4\}$ be the set of feed valves with respect to the quenching streams (i.e., Streams 1-4 in Figure 1), m_f denote the mass flowrate of the raw feed stream, $\tilde{m}_v$ denote the mass flowrate through valve v , and $\tilde{r}_v$ be the split ratio of valve v . The mass balance is given by:

$$(m_f + m_{cyc})\tilde{r}_v = \tilde{m}_v, \quad \forall v \in \mathcal{V} \quad (34)$$

$$\sum_{v \in \mathcal{V}} \tilde{r}_v = 1 \quad (35)$$

It is assumed that the feed stream and the recycle stream are at the same temperature denoted by T_f . The feed stream to the first reactor bed is a mixture of the second quenching stream (Stream 2 in Figure 1) and the output of HE-1. The mass and heat balances of the mixing process are given by:

$$m_1 = \tilde{m}_1 + \tilde{m}_2 \quad (36)$$

$$m_1 C_{p11} T_{in1} = \tilde{m}_1 \tilde{C}_p T_{HEC,1} + \tilde{m}_2 \tilde{C}_p T_f \quad (37)$$

where $\tilde{C}_p$ is the specific heat capacity of the total feed stream (raw feed stream mixed with the recycle stream). Similarly, for the second and the third reactor bed, the following equations hold:

$$m_2 = \tilde{m}_3 + m_1 \quad (38)$$

$$m_3 = \tilde{m}_4 + m_2 \quad (39)$$

$$m_2 C_{p21} T_{in2} = m_1 C_{p1n} T_{out1} + \tilde{m}_3 \tilde{C}_p T_f \quad (40)$$

$$m_3 C_{p31} T_{in3} = m_2 C_{p2n} T_{out2} + \tilde{m}_4 \tilde{C}_p T_f \quad (41)$$

Let T_{in} and T_{out} denote the inlet and outlet temperatures of the reactor beds, respectively. The relations between T_{in} , T_{out} and T_g are as follows:

$$T_{in_i} = T_{g_{i0}}, \quad \forall i \in I \quad (42)$$

$$T_{out_i} = T_{g_{in}}, \quad \forall i \in I \quad (43)$$

3. Control Problem

In the operation of the ammonia synthesis plant, drastic oscillations in the reactor temperatures and overheating of the catalyst beds are undesirable as they may lead to detrimental catalyst deactivation (Rosbo et al., 2023). Also, frequent and drastic pressure changes in the reactor beds or the separator should be avoided, as they can cause fatigue stress to the vessels (Ostuni and Zardi, 2016). Since it is assumed that compressor C2 can always pressurize the feed stream to the desired pressure, the feed pressure is fixed at the nominal value (200 bar), thus the main control objectives are to control the reactor temperatures and the pressure in the flash tank. The liquid volume in the flash tank should also be controlled within the specified safety range (stated in Section 2).

Based on the process flowsheet in Figure 1, there are seven adjustable valves: four at the feed streams to the reactor beds and the heat exchanger, one at the bottom of the flash tank, one at the gas phase of the flash tank, and one at the purge splitter. It is natural to consider the corresponding variables, namely, the

split ratios of the four feed valves, the gas phase output flowrate of the flash tank, the purge flowrate, and the production rate as the manipulated inputs $\mathbf{u}$:

$$\mathbf{u} = [\tilde{r} \in \mathbb{R}^{|\mathcal{V}|} \quad F_v \quad F_p \quad F_l]^\top \quad (44)$$

Some discussion on the choice of controlled variables can be found in Zhang et al. (2010) and Araújo and Skogestad (2008), in which the feed temperatures of reactor beds, and the pressure and liquid volume in the separator are controlled. In our case, as no inert component (CH_4 or Ar) is contained in the synthesis gas, the recycle flowrate should also be controlled as close to the vapor outlet rate from the separator as possible to reduce the loss of synthesis gas. Moreover, when the reactor load changes significantly, the temperature along the whole reactor bed may not be stabilized by only controlling the feed temperature, thus, the end temperatures are also controlled. In summary, the controlled variables include: feed and end temperatures of the reactor beds (six in total), pressure in the separator, liquid volume in the separator, and the recycle flowrate. The vector of controlled variables $\hat{\mathbf{x}}$ is then as follows:

$$\hat{\mathbf{x}} = [T_{in} \in \mathbb{R}^{|I|} \quad T_{out} \in \mathbb{R}^{|I|} \quad P_{sep} \quad V_l \quad F_{cyc}]^\top \quad (45)$$

The set point of the flash pressure is the nominal operating condition (196 bar), while the set points of the reactor temperatures may vary with different reactor loads, thus, the set points for these temperatures are obtained by performing a steady-state optimization that maximizes the ammonia production rate:

$$\underset{\mathbf{x}, \mathbf{u}}{\text{maximize}} \quad F_l x_{NH3} \quad (46)$$

$$\text{subject to} \quad Eqs. 1 - 43 \quad (47)$$

$$\frac{dT_{c_{ij}}}{dt} = 0, \quad \forall i \in I, j \in \mathcal{J} \quad (48)$$

$$\frac{dT_{g_{ij}}}{dt} = 0, \quad \forall i \in I, j \in \mathcal{J} \quad (49)$$

$$\frac{dN_s}{dt} = 0, \quad \forall s \in \mathcal{S} \quad (50)$$

$$\bar{\mathbf{x}}_{min} \leq \mathbf{x} \leq \bar{\mathbf{x}}_{max} \quad (51)$$

$$\bar{\mathbf{u}}_{min} \leq \mathbf{u} \leq \bar{\mathbf{u}}_{max} \quad (52)$$

where $\mathbf{x}$ represent state variables, $\bar{\mathbf{x}}_{min}$ and $\bar{\mathbf{x}}_{max}$ are the lower and upper bounds of $\mathbf{x}$, and $\bar{\mathbf{u}}_{min}$ and $\bar{\mathbf{u}}_{max}$ are the lower and upper bounds of $\mathbf{u}$, respectively. Detailed information on $\mathbf{x}$ and the corresponding bounds is shown in Table 3; the values of $\bar{\mathbf{u}}_{min}$ and $\bar{\mathbf{u}}_{max}$ are shown in Table 4.

To further test the robustness of the control structure, we add disturbances $w(t)$ to the nominal reaction rate $r_{NH3}(t)$, which are sampled from a Gaussian distribution:

$$\tilde{r}_{NH3}(t) = r_{NH3}(t) + w(t), \quad w(t) \sim \mathcal{N}(0, \sigma), \quad \forall t \in \mathcal{T} \quad (53)$$

where $\mathcal{T}$ is the total time horizon. Different disturbance levels are considered by using different values of σ .

4. NMPC Formulation

The overall control scheme is shown in Figure 2. The NMPC takes inputs of real-time measurements of the system states $\mathbf{x}_{init}$

and the control actions $\mathbf{u}_{init}$ from the ammonia synthesis plant, and the set points $\mathbf{x}^{sp}$ from the steady-state optimization layer (Eqs. 46 - 52). It is assumed that all state variables of the ammonia plant can be measured or accurately reconstructed through a state observer.

The NMPC provides optimal control actions $\mathbf{u}^*(t_k)$ by solving a dynamic optimization problem involving a set of differential-algebraic equations (DAEs) with a prediction time horizon of $\mathcal{H} = [t_0, t_f]$, where f represents the number of discrete time segments and t_k represents the k th time instance. The dynamic optimization model is formed as follows:

$$\underset{\mathbf{u}}{\text{minimize}} \quad \sum_{k=1}^f L(\hat{\mathbf{x}}(t_k), \mathbf{x}^{sp}, \mathbf{u}(t_k), \mathbf{u}(t_{k-1})) \quad (54)$$

$$\text{subject to} \quad Eqs. 1 - 43 \quad (55)$$

$$|\mathbf{x}(t_k) - \mathbf{x}(t_{k-1})| \leq \mathbf{C}_x, \quad \forall k = 1, 2, \dots, f \quad (56)$$

$$\mathbf{C}_u^- \leq \frac{\mathbf{u}(t_k) - \mathbf{u}(t_{k-1})}{\mathbf{u}(t_{k-1})} \leq \mathbf{C}_u^+, \quad \forall k = 1, 2, \dots, f \quad (57)$$

$$\bar{\mathbf{x}}_{min} \leq \mathbf{x}(t) \leq \bar{\mathbf{x}}_{max} \quad (58)$$

$$\bar{\mathbf{u}}_{min} \leq \mathbf{u}(t) \leq \bar{\mathbf{u}}_{max} \quad (59)$$

$$\mathbf{x}(0) = \mathbf{x}_{init} \quad (60)$$

$$\mathbf{u}(0) = \mathbf{u}_{init} \quad (61)$$

$$V_l(t_f) = 0.3V_{sep} \quad (62)$$

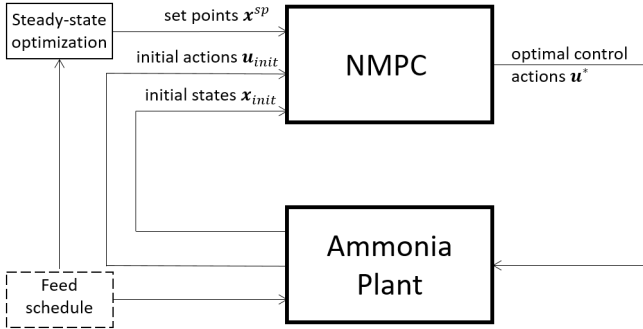


Figure 2: The proposed control scheme.

4.1. Objective Function

The objective function in Eq. 54 is the sum of the loss functions over the prediction time horizon. The loss function $L(\cdot)$ is formulated as follows:

$$L = (\hat{\mathbf{x}} - \mathbf{x}^{sp})^\top \mathbf{Q}_x (\hat{\mathbf{x}} - \mathbf{x}^{sp}) + (\mathbf{u}(t_k) - \mathbf{u}(t_{k-1}))^\top \mathbf{Q}_u (\mathbf{u}(t_k) - \mathbf{u}(t_{k-1})), \quad \forall k = 1, 2, \dots, f \quad (63)$$

in which the first term corresponds to the deviations of the controlled variables from the set points, and the second term corresponds to the penalties on the manipulated inputs. The weight matrices $\mathbf{Q}_x$ and $\mathbf{Q}_u$ are positive semi-definite and diagonal, and they are tuned in the simulation case study in Section 5 to reduce offsets as much as possible while meeting all constraints. The entries in $\mathbf{Q}_x$ and $\mathbf{Q}_u$ corresponding to each controlled variable or manipulated input are shown in Table 5.

Table 3: Bounds for state variables

Variable	Domain	$\bar{\mathbf{x}}_{min}$	$\bar{\mathbf{x}}_{max}$	Unit
m	$\mathbb{R}^{ \mathcal{I} }$	0	200	kg/s
$\tilde{m}$	$\mathbb{R}^{ \mathcal{V} }$	0	200	kg/s
Tg	$\mathbb{R}^{ \mathcal{I} \times \mathcal{J} }$	500	900	K
Tc	$\mathbb{R}^{ \mathcal{I} \times \mathcal{J} }$	500	900	K
r	$\mathbb{R}^{ \mathcal{I} \times \mathcal{J} }$	-1×10^4	1×10^4	$\frac{\text{kgNH}_3}{\text{kgCat.s}}$
Cp	$\mathbb{R}^{ \mathcal{I} \times \mathcal{J} }$	0	2×10^4	J/kg/K
P	$\mathbb{R}^{ \mathcal{S} \times \mathcal{I} \times \mathcal{J} }$	0	200	bar
w	$\mathbb{R}^{ \mathcal{S} \times \mathcal{I} \times \mathcal{J} }$	0	1	N/A
$T_{HEC,0}$	$\mathbb{R}$	300	600	K
$T_{HEC,1}$	$\mathbb{R}$	500	800	K
$T_{HEH,0}$	$\mathbb{R}$	500	900	K
$T_{HEH,1}$	$\mathbb{R}$	500	900	K
P_{sep}	$\mathbb{R}$	190	200	bar
V_l	$\mathbb{R}$	$0.25V_{sep}$	$0.4V_{sep}$	m^3
F_v	$\mathbb{R}$	0	3000	mol/s
N	$\mathbb{R}^{ \mathcal{S} }$	0	4×10^6	mol
N_l	$\mathbb{R}$	0	4×10^6	mol
N_v	$\mathbb{R}$	0	1×10^6	mol
x	$\mathbb{R}^{ \mathcal{S} }$	0	1	N/A
y	$\mathbb{R}^{ \mathcal{S} }$	0	1	N/A

Table 4: Bounds for manipulated inputs

Variable in $\mathbf{u}$	Domain	$\bar{\mathbf{u}}_{min}$	$\bar{\mathbf{u}}_{max}$	Unit
$\tilde{r}$	$\mathbb{R}^{ \mathcal{V} }$	0	1	N/A
F_v	$\mathbb{R}$	60	3000	mol/s
F_{cyc}	$\mathbb{R}$	60	3000	mol/s
F_l	$\mathbb{R}$	0	500	mol/s

4.2. Ramping Constraints

Ramping constraints on the state variables (Eq. 56) and the manipulated inputs (Eq. 57) are also considered. The parameters $\mathbf{C}_x$ are bounds for the ramping of state variables; $\mathbf{C}_u^-$ and $\mathbf{C}_u^+$ are lower and upper bounds for the ramping of manipulated inputs, respectively. For the controlled variables, constraints aiming to prevent drastic temperature/pressure changes are as follows:

$$|Tg_{ij}(t_k) - Tg_{ij}(t_{k-1})| \leq 5, \quad \forall k = 1, 2, \dots, f, \quad i \in \mathcal{I}, \quad j \in \mathcal{J} \quad (64)$$

$$|Tc_{ij}(t_k) - Tc_{ij}(t_{k-1})| \leq 5, \quad \forall k = 1, 2, \dots, f, \quad i \in \mathcal{I}, \quad j \in \mathcal{J} \quad (65)$$

$$|P_{sep}(t_k) - P_{sep}(t_{k-1})| \leq 1, \quad \forall k = 1, 2, \dots, f \quad (66)$$

Eq. 64 and Eq. 65 state that the temperature changes of the synthesis gas and catalyst particles in reactor beds in neighbouring time instances can not be greater than 5 K; Eq. 66 states that the pressure change in the separator in adjacent time instances can not be greater than 1 bar.

Among the manipulated inputs, it is intuitive to have faster adjustment of the split ratios of the quenching streams to stabilize the reactor temperature, whereas the vapor-phase outlet flowrate of the flash tank and the purge flowrate should not

Table 5: Controlled variables, set points, manipulated inputs, and corresponding weights in the loss function

Variable in $\hat{\mathbf{x}}$	$\mathbf{x}^{sp}$	Weight in Q_x
$T_{in_i}, \forall i \in \mathcal{I}$	from optimization	0.6
$T_{out_i}, \forall i \in \mathcal{I}$	from optimization	0.25
P_{sep}	196	1
V_l	$0.3 V_{sep}$	10
F_{cyc}	F_v	0.001
Variable in $\mathbf{u}$		Weight in Q_u
F_l		0.01
F_p		5×10^{-3}
F_v		0
$\tilde{r}$		0

change equally fast as it can cause drastic temperature and pressure oscillations. Also, the production rate ramping should be constrained to ensure smooth transitions and reduce fluctuations. Thus it is assumed that the split ratios ($\tilde{r}$) of the quenching streams at the next time instance can be 80% - 120% of the current value, whereas the remaining manipulated variables are only allowed to vary between 90% and 110% of their current values:

$$0.8 \leq \frac{\tilde{r}_v(t_k)}{\tilde{r}_v(t_{k-1})} \leq 1.2, \quad \forall k = 1, 2, \dots, f, \quad v \in \mathcal{V} \quad (67)$$

$$0.9 \leq \frac{F_l(t_k)}{F_l(t_{k-1})} \leq 1.1, \quad \forall k = 1, 2, \dots, f \quad (68)$$

$$0.9 \leq \frac{F_p(t_k)}{F_p(t_{k-1})} \leq 1.1, \quad \forall k = 1, 2, \dots, f \quad (69)$$

$$0.9 \leq \frac{F_v(t_k)}{F_v(t_{k-1})} \leq 1.1, \quad \forall k = 1, 2, \dots, f \quad (70)$$

4.3. Initial and Terminal Conditions

Initial conditions for state variables and control actions are stated in Eqs. 60 and 61, respectively. Initial values $\mathbf{x}_{init}$ and $\mathbf{u}_{init}$ are obtained from real-time measurements at the ammonia plant, and are assigned to $\mathbf{x}(0)$ and $\mathbf{u}(0)$. Furthermore, as shown in Eq. 62, a terminal condition is applied to the liquid volume in the separator, which is the nominal volume defined for the process.

5. Simulation

5.1. Simulation Setup

A simulation of the flexible ammonia synthesis process described in Section 2 with the NMPC scheme formulated in Section 4 was implemented. Designs, operating conditions, and catalyst properties are provided in Tables 1-2, and any other parameters are shown in Table B.9 in the Appendix B.

Given the full-load feed flowrate of 600 mol/s, an assumed feed schedule is set for a 5-hour time horizon, in which the feed flowrate varies once per hour: the feed flowrate starts at 304 mol/s, increases to 504 mol/s in the next hour, then decreases to 477 mol/s and 316 mol/s in the next two step changes, and it

Table 6: Feed flowrate profile and the respective percentage of the nominal load.

Time duration	Flowrate, mol/s	Reactor load
Hour 0 - 1	304	50.5%
Hour 1 - 2	504	84.0%
Hour 2 - 3	477	79.5%
Hour 3 - 4	316	52.7%
Hour 4 - 5	572	95.3%

finally increases to 572 mol/s. The feed profile is provided in Table 6.

As stated in Section 3, Gaussian disturbances are added to the reaction rates. Three different values are considered for the standard deviation σ in Eq. 53: $\sigma_1 = 0$ (disturbance-free), $\sigma_2 = 0.1 r_{NH_3}$, and $\sigma_3 = 0.2 r_{NH_3}$.

The NMPC is implemented with the Python package Pyomo (Hart et al., 2011; Bynum et al., 2021). The source codes of the NMPC scheme, along with the simulation model are available on Github: <https://github.com/ddolab/AmmoniaSynCtrl>. The prediction time horizon is set to be 180 seconds, and the Backward Euler method is used to discretize the model in time with a 10-second discretization step, based on which $f = 180/10 = 18$. The resulting nonlinear program (NLP) that consists of 7638 variables among which 1134 are differential state variables and the remainder are algebraic variables, 7525 equality constraints, and 2488 inequality constraints is solved by IPOPT (Wächter and Biegler, 2006). The variables are initialized with values from the previous prediction result or from the steady-state optimization. Note that when an optimal solution to the prediction problem is not obtained, the control inputs returned by the solver are still adopted.

5.2. NMPC Results

The temperature profiles at the inlet and outlet of each reactor bed are shown in Figures 3 and 4 as solid lines, whereas the dashed lines represent the corresponding set points. All temperatures are well maintained within a safe operating range (i.e., between 600 K and 900 K), ensuring that the reactor beds neither cool down excessively nor experience catalyst deactivation. Optimal temperatures (set points) vary depending on the feed flowrate, with the maximum increase in a single step being approximately 45 K. The controller is able to bring the temperatures close to the corresponding set points smoothly within approximately 10 minutes following a step change in the feed flowrate, though small offsets exist. The offsets are most likely due to the different discretizations of the ODEs when solving the prediction NLP problem and simulating the plant. Comparing across the three scenarios with different disturbance levels, slight temperature fluctuations are observed when the standard deviation increases from 0 to σ_2 , and the fluctuations are significantly larger when the standard deviation is set to σ_3 . Despite these fluctuations, the inlet and outlet temperatures are still controlled near their set points in all the cases.

Figure 5 shows the profile of the pressure in the separator, along with the recycle flowrate and purge flowrate that have

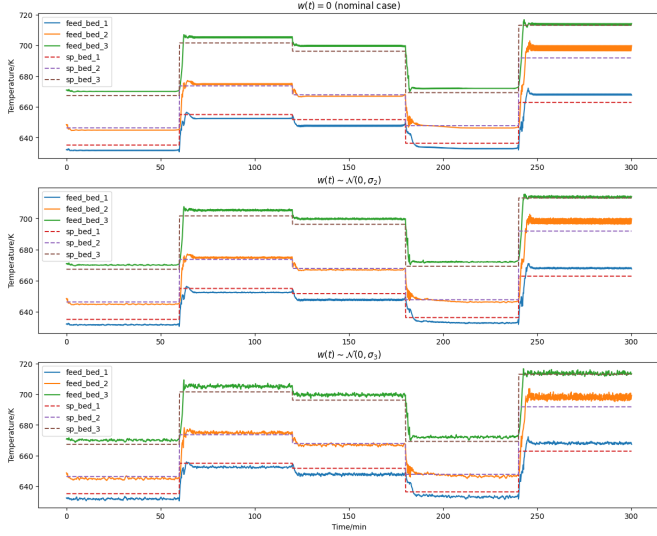


Figure 3: The inlet temperatures profiles of the reactor beds.

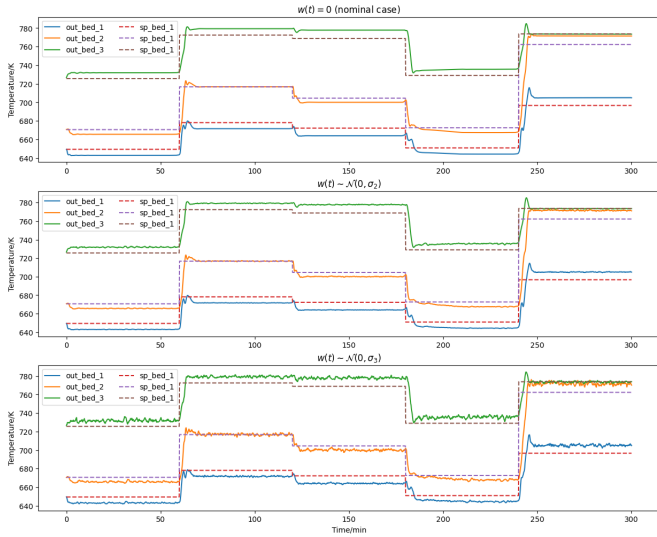


Figure 4: The outlet temperatures profiles of the reactor beds.

significant influence on the pressure. The pressure is well maintained within the desired operating range (190-200 bar) and is controlled near the nominal operating pressure of 196 bar. The pressure increases when the feed flowrate ramps up, and it drops when the feed flowrate ramps down. It takes up to 40 minutes for the pressure to reach steady state after a step change in the reactor load, which is slower than the temperature transitions. This is due to the relatively strict constraints set for the ramping of the recycle flowrate to avoid drastic pressure and temperature fluctuations. It is also evident that the purge flowrate increases to reduce the pressure in the separator when it gets close to the upper bound, and is maintained near zero when the plant is at steady state. When the disturbances increase, larger fluctuations in the pressure are observed, resulting in more frequent adjustments of the recycle flowrate to stabilize the pressure. Still, the offset of the pressure is not completely eliminated.

The ammonia production rate and the ammonia concentra-

tion in the product stream are shown in Figure 6. It is interesting to note that even though the production rate is not directly controlled, by controlling other state variables such as temperatures in the reactor beds and the pressure of the separator, the actual production rate stays very close to the optimal production rate obtained from the steady-state optimization. No fluctuation of the production rate is observed when the disturbance increases, as it is directly manipulated. When the feed flowrate changes, the transition between production rates takes up to 20 minutes. The transitions are smooth and the production rate profile is steady after the transitions, due to the penalties on the ramping of the production rate. The mole fraction of ammonia in the product varies slightly around the nominal concentration 0.9765, and is in the range of 0.9760 – 0.9768, depending on the separation pressure. The quality of the product, i.e. the ammonia concentration, is acceptable considering the reference values of 0.96 and 0.99 reported in Zhang et al. (2010) and Lin et al. (2020), respectively. Minor fluctuations appear in the ammonia concentration when the disturbance increases. Transitions of the ammonia concentration between steady states are slower than that of the production rate, due to the slow pressure changes in the flash tank.

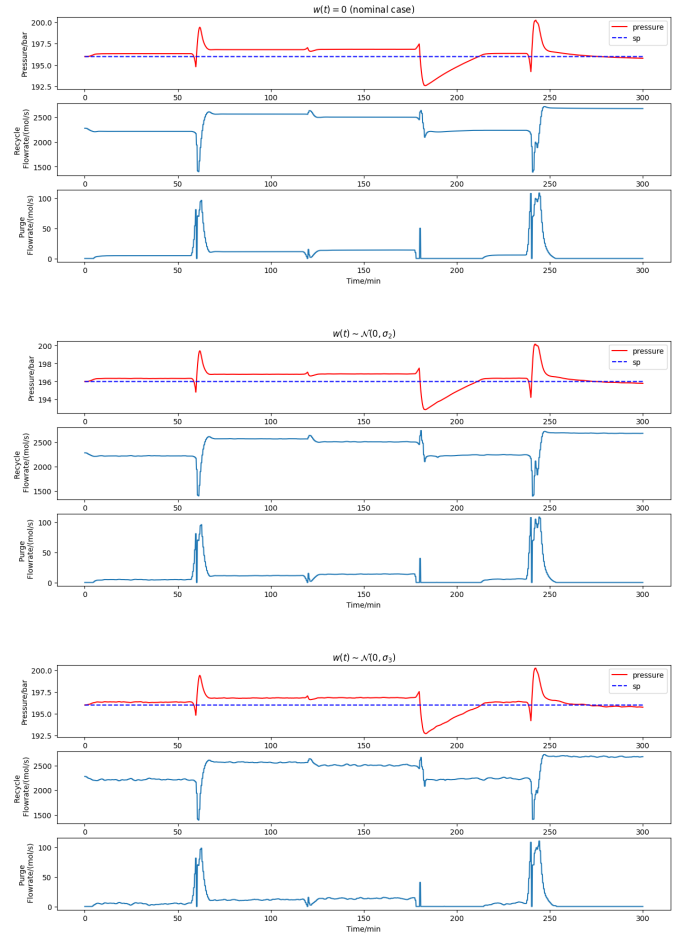


Figure 5: The profiles of separation pressure, recycle flowrate, and purge flowrate.

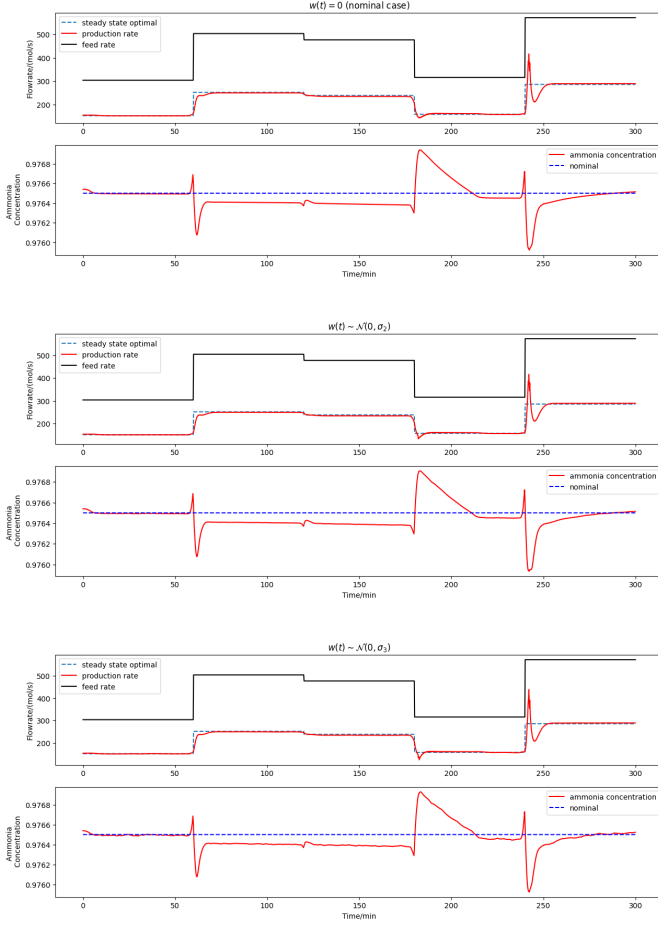


Figure 6: The profiles of production rate and ammonia concentration.

5.3. Comparison with PID Control

To further evaluate the control performance of the proposed NMPC scheme, a PID control structure adopted from Araújo and Skogestad (2008); Zhang et al. (2010) is implemented as a reference. The PID control structure is shown in Figure 7, where PI controllers are deployed to track feed temperatures of reactor beds and the pressure in the separator, while a P controller is used to regulate liquid volume in the flash tank. The purge valve is opened if the actual vapor output flowrate of the separator can not be increased to the value specified by the controller in time, due to the ramping constraints applied to the recycle valve. The PID scheme is implemented in Simulink 2024a, where controller parameters are tuned via the built-in tuning functionality. Details of the controlled variables with the corresponding manipulated inputs and controller parameters are shown in Table 7.

The PID control structure is first tested with the feed flowrate profile shown in Table 6; however, the separation pressure and the temperature of reactors can not be stabilized in one hour after a drastic change of the feed flowrate. Thus, the time periods for each feed flowrate (except the first one) are extended from 1 hour to 1.5 hour, as shown in Table 8. Other parameters such as equipment sizes, catalyst properties, and initial temperatures

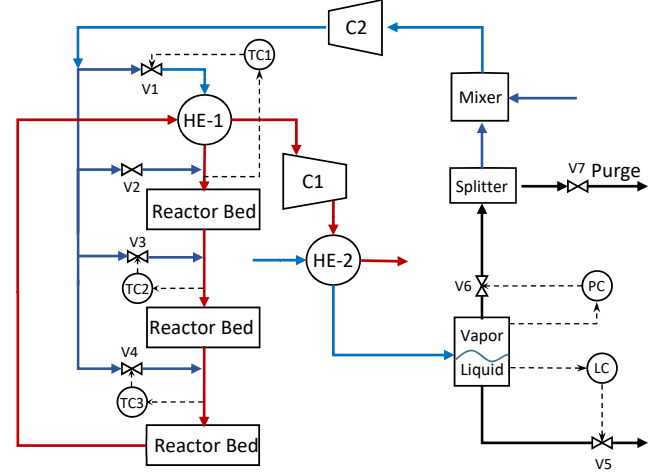


Figure 7: The PID control structure.

and pressures are consistent with those described in Section 2. To make a fair comparison, the NMPC scheme is also tested with the same feed flowrate profile. Disturbances in the reaction rates are not considered.

The comparison of feed temperatures of the reactor beds is shown in Figure 8. The PI controllers manage to track the set points with zero offsets; however, significant oscillations that can be detrimental to the catalysts are observed, especially after a large step increase of the feed flowrate at the 330th minute. In contrast, the NMPC results show no or little temperature oscillations despite the small offsets, and the temperatures reach steady state at least 3 times faster than the those obtained from the PID control scheme. Separation pressures in the NMPC result and the PID control result are shown in Figure 9. The PI controller tracks the reference pressure in 60 - 80 minutes after a large step change in the feed flowrate, which is significantly longer than the time required by the NMPC. The production flowrate and ammonia concentration in the product are shown in Figure 10. Both control schemes adjust the production flowrate as a manipulated variable which stays at the optimal value after a steady state is reached. However, in the PID control, more fluctuations are observed in the production flowrate profile and it takes longer to have a steady product output after a 43% feed flowrate increase at the 330th minute. The ammonia purity in the product is acceptable in both cases, while in the NMPC it exhibits less variance due to faster tracking of the separation pressure. Overall, the NMPC scheme outperforms the PID control scheme in terms of the temperature/pressure tracking speed and the transition smoothness, though minor offsets (less than 1% of the set points) may exist.

6. Conclusions

In this work, an NMPC scheme for the control of a flexible ammonia production system is proposed and tested. A feed schedule in which the feed flowrate varies between 50% and 100% of the nominal value is assumed for a 5-hour time horizon. The simulations demonstrate the feasibility of the pro-

Table 7: Configuration of the PID control structure

Controller	Type	Controlled Variable	Manipulated Inputs	Parameter
TC1	PI	T_{in_1}	Opening of V1	$P = -0.04886, I = -0.003$
TC2	PI	T_{in_2}	Opening of V3	$P = -0.04391, I = -0.057$
TC3	PI	T_{in_3}	Opening of V4	$P = -0.02123, I = -0.0008$
LC	P	V_l	Opening of V5	$P = -2 \times 10^{-5}$
PC	PI	P_{sep}	Opening of V6	$P = -0.254, I = -0.0003$

Table 8: Modified feed flowrate profile for the comparison case study.

Time duration	Flowrate, mol/s	Reactor load
Hour 0 - 1.0	304	50.5%
Hour 1.0 - 2.5	504	84.0%
Hour 2.5 - 4.0	477	79.5%
Hour 4.0 - 5.5	316	52.7%
Hour 5.5 - 7.0	572	95.3%

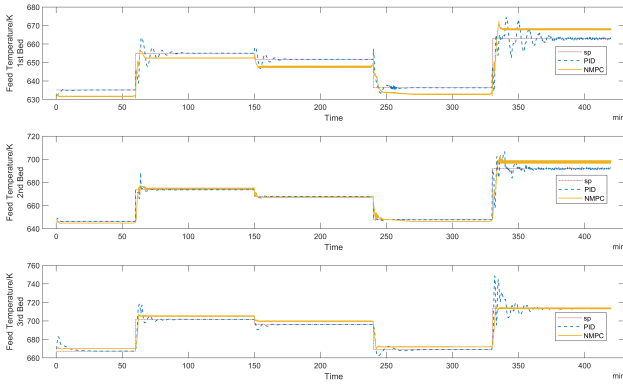


Figure 8: The feed temperature profiles in the results of the PID control and NMPC schemes.

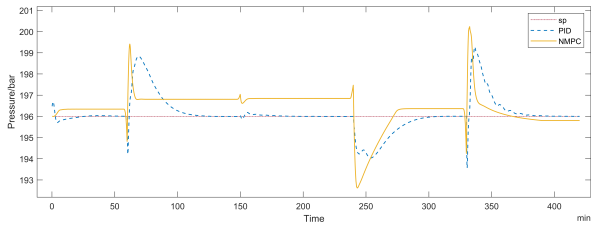


Figure 9: The separation pressure profiles in the results of the PID control and NMPC schemes.

posed control scheme. Results show that all the controlled variables are maintained in a safe operating range and tracked near the corresponding set points, with small offsets. The transitions of reactor temperatures take around 10 minutes, and it takes around 40 minutes for the pressure in the flash tank to reach a steady state for a large step change in the feed flowrate. When normally distributed disturbances are introduced to the reaction rate, minor fluctuations are observed in both the profiles of the reactor temperature and the separation pressure. The production rate, on the other hand, is not affected by these disturbances as it is directly manipulated, and it stays close to the optimal values dictated by the steady-state optimization. The ammonia

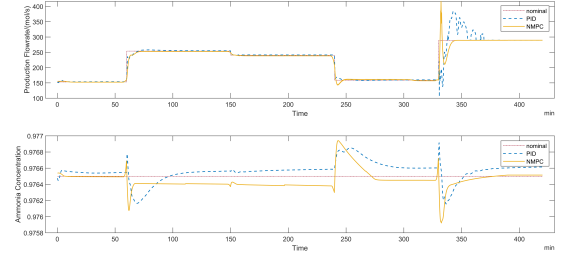


Figure 10: The production flowrate and ammonia concentration in the results of the PID control and NMPC schemes.

concentration in the product is between 97.60% and 97.68%, which is close to the nominal value of 97.65%. Compared to a PID control scheme, the NMPC tracks the temperatures and the pressure significantly faster, and a smoother transition in the production flowrate is also observed.

In all, the proposed NMPC strategy is demonstrated to work well for the considered flexible ammonia synthesis process, under the assumption that all variables are measured accurately and are noise-free. Future research efforts will focus on addressing the control problem under realistic measurements of the state variables and addressing the control of the process over a wider operating range (e.g. 10%-100% reactor load variation).

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Appendix A. Nomenclature in the NMPC formulation

Sets

- $\mathcal{H} = [t_0, t_f]$, the prediction time horizon
- $\mathcal{V} = \{1, 2, 3, 4\}$, the set of feed valves
- $\mathcal{I} = \{1, 2, 3\}$, the set of reactor beds
- $\mathcal{J} = \{1, 2, \dots, N\}$, the set of discrete reactor sections
- $\mathcal{S} = \{H_2, N_2, NH_3\}$, the set of chemical species

Variables

- F_v , gas phase output flowrate of the separator
- F_{cyc} , recycle molar flowrate

m_{cyc} , recycle mass flowrate
 m_f , feed mass flowrate
 F_l , liquid phase output flowrate of the separator (production rate)
 F_p , purge flowrate
 P_{sep} , pressure in the separator
 V_l , liquid volume in the separator
 C_c , heat capacity rate of the cold stream fed to HE-1
 C_h , heat capacity rate of the hot stream fed to HE-1
 $T_{HEC,0}$, temperature of the cold stream fed to HE-1
 $T_{HEC,1}$, temperature of the cold stream leaving HE-1
 $T_{HEH,0}$, temperature of the hot stream fed to HE-1
 $T_{HEH,1}$, temperature of the hot stream leaving HE-1
 q , amount of heat transferred in HE-1
 C_r , heat capacity ratio
 ϵ , effectiveness of HE-1
 $\tilde{r}_v$, $v \in \mathcal{V}$, split ratios of the quench stream through valve v
 $\tilde{m}_v$, $v \in \mathcal{V}$, mass flowrate through valve v
 $T_{in,i}$, $i \in \mathcal{I}$, inlet temperature of reactor i
 $T_{out,i}$, $i \in \mathcal{I}$, outlet temperature of reactor i
 m_i , $i \in \mathcal{I}$, feed mass flowrate of reactor i
 N_s , $s \in \mathcal{S}$, holdup of species s in the separator
 y_s , $s \in \mathcal{S}$, vapor mole fraction of species s in the separator
 x_s , $s \in \mathcal{S}$, liquid mole fraction of species s in the separator
 T_{gij} , $i \in \mathcal{I}$, $j \in \mathcal{J}$, fluid temperature at section j in reactor i
 T_{cij} , $i \in \mathcal{I}$, $j \in \mathcal{J}$, catalyst bed temperature at section j in reactor i
 r_{ij} , $i \in \mathcal{I}$, $j \in \mathcal{J}$, the reaction rate at section j in reactor i
 $\dot{Q}_{ij}$, $i \in \mathcal{I}$, $j \in \mathcal{J}$, heat transfer rate between the fluid and catalyst at section j in reactor i
 Cp_{ij} , $i \in \mathcal{I}$, $j \in \mathcal{J}$, specific heat capacity of the fluid at section j in reactor i
 Pr_{ij} , $i \in \mathcal{I}$, $j \in \mathcal{J}$, Prandtl number the fluid at section j in reactor i
 Re_{ij} , $i \in \mathcal{I}$, $j \in \mathcal{J}$, Reynolds number the fluid at section j in reactor i
 Nu_{ij} , $i \in \mathcal{I}$, $j \in \mathcal{J}$, Nusselt number the fluid at section j in reactor i
 P_{sij} , $s \in \mathcal{S}$, $i \in \mathcal{I}$, $j \in \mathcal{J}$, partial pressure of species s at section j in reactor i
 z_{sij} , $s \in \mathcal{S}$, $i \in \mathcal{I}$, $j \in \mathcal{J}$, mole fraction of species s at section j in reactor i
 w_{sij} , $s \in \mathcal{S}$, $i \in \mathcal{I}$, $j \in \mathcal{J}$, molecular weight of species s at section j in reactor i

Appendix B. Parameters in Simulation

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Table B.9: Parameter values used in the simulation.

Parameter	Unit	Value
n	N/A	10
f	N/A	18
R	J/mol/K	8.314
μ	Pa·s	0.028
λ	W/m/K	0.231
ΔH_r	kJ/kg	-2900
ΔT	K	20
MW_{H_2}	kg/kmol	2.016
MW_{N_2}	kg/kmol	28.02
MW_{NH_3}	kg/kmol	17.03
$\tilde{C}p_{H_2}$	kJ/kg/K	14.6
$\tilde{C}p_{N_2}$	kJ/kg/K	1.1
$\tilde{C}p_{NH_3}$	kJ/kg/K	3.16
$\rho_{l_{H_2}}$	kg/m ³	70.85
$\rho_{l_{N_2}}$	kg/m ³	807
$\rho_{l_{NH_3}}$	kg/m ³	561
$P_{NH_3}^{sat}$	bar	15.40
H_{H_2}	bar	6838
H_{N_2}	bar	11032

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